UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER 2022 EXAMINATION

UECS3253 WIRELESS APPLICATION DEVELOPMENT

MONDAY, 26 SEPTEMBER 2022 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF SCIENCE (HONOURS) SOFTWARE ENGINEERING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and ONLY **ONE (1)** question in **Section B**.

Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in **Section B**, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

The use of a calculator is prohibited.

UECS3253 WIRELESS APPLICATION DEVELOPMENT

Section A: Answer **ALL** questions.

Q1. (a) Identify **THREE (3)** different types of mobile apps and give description to each of them. (12 marks)

(b) React Native provides a number of built-in components. Those components divides the user interface into reusable, independent parts that can be processed independently. Give **FOUR (4)** examples of React Native components and describe their usage. (8 marks)

(c) Describe Flexbox in React Native and give **TWO (2)** of its common used properties. (5 marks)

[Total : 25 marks]

Q2. **Company X** is selling healthcare products and would like to create a mobile application with a native database to store the data of any healthcare products on sale. A pre-populated SQLite database named **productSalesDB.sqlite** is created, with a table named **SalesProducts_Table**, recorded the data of any healthcare products on sale as shown in Table 1. At the beginning of each month, **Company X** will compile a list of the products on sale and upload it to the company website's cloud storage. The mobile application will automatically download the list to be used for data updates. Figure 1 depicts the example of the downloaded product list, which is declared as a variable named **product_List** in JSON format.

id	name	price
1	S'clean-Facemask	RM15
2	Slim-vitaminC	RM35
3	Whitening Cream	RM125

Table 1: **SalesProducts_Table**

```
[
  {"id" : "5",
   "name" : "Super-Fitness Gear",
   "price": "RM111"
  },

  {"id" : "6",
   "name" : "F-weight-loss Gel",
   "price": "RM31"
  },

  {"id" : "7",
   "name" : "G-bad breath spray",
   "price": "RM19"
  },
]
```

Figure 1: **product_List**

Q2. (Continued)

(a) Construct code segment to QUERY whether a particular product in **product_List** has already existed in **SalesProducts_Table**. If it is NOT, store the product details by INSERT a new row to **SalesProducts**. (10 marks)

(b) Construct code segment to QUERY whether a particular product in **product_List** has already existed in **SalesProducts_Table**. If it is YES, update the product details by their ID. (10 marks)

(c) Create a code segment to render each product in **product_List** as a FlatList item which showing the item's name and price. (5 marks)

[Total : 25 marks]

Q3. (a) Describe AsyncStorage in React Native and give **ONE (1)** advantage and **ONE (1)** disadvantage of using AsyncStorage. (7 marks)

(b) Table 2 lists the available options, as well as their names and preference values. Create a code segment that save all the names and the corresponding preference values into AsyncStorage. (9 marks)

```
this.state=
{ playSound : false,
showNotification :false,
language : 'English',
showNearby :false,
shareData: true
Wifi_on : true}
```

Table 2

(c) Table 3 lists the available options that have saved, as well as their key names and sample values. Create a code segment that loads the values of all keys in Table 3 and sets the states accordingly. (9 marks)

Key	Sample Value
playSound	false
showNotification	false
language	English
showNearby	false
shareData	true
Wifi_on	true

Table 3

[Total : 25 marks]

Section B: Answer ONLY **ONE (1)** question.

Q4. AJ Hair Studio has an online booking platform where customers can schedule haircut appointments. With the help of a mobile app, users can schedule an appointment with their favourite barber.

(a) To make an online reservation, a customer must first register in a database maintained by a web server. Figure 2.1 depicts the JSON-formatted user sample data sent to the web server by the app. In exchange, the web server will assign each registered user an ID along with a 10% discount voucher code. Figure 2.2 depicts a sample response data sent by the web to the mobile app in JSON format.

Create a code segment to send the JSON data in Figure 2.1 to a web server via the HTTP POST method at the URL **https://ajhairstudio/api/register**. Assume the JSON data is already assigned to a variable of state called **user**.

If the user is successfully registered, navigate the user to a screen named **Login** and pass the `user_ID` and `voucher_CODE` as a route parameters. Otherwise, an alert with an appropriate error message should be displayed.

If an error occurs while attempting to connect to and communicate with the web server, an error message should be displayed in the console. (13 marks)

```
{
"name":"Yong Kok Jian",
"gender": "Female",
"email": "YKJ@hotmail.com",
"phone": "+60123456789" ,
"password": "yjkasd123",
}
```

Figure 2.1

```
{
"user_ID":"1011",
"voucher_CODE": "Xcs23",
"success ": true,}
```

Figure 2.2

(b) After logging into the mobile app, the user shall make a query for any available time slots that include the date and time of a specific preferred barber. As shown in Figure 3.1, the web server will send a response to the mobile app informing the user of a specific preferred barber's available time slots.

Create a code segment for the above query and submit it to the web server using the HTTP GET method at **https://ajhairstudio/api/register.**

Q4. (b) (Continued)

If there is any available time slots, navigate the user to another screen named **Start_booking** and pass all available time slots as route parameters. Otherwise, an alert with an appropriate error message should be displayed.

If an error occurs while attempting to connect to and communicate with the web server, an alert with an appropriate error message should be displayed.

(12 marks)

```
{ "time_slot_1":{"date":
"10/05/2021",
"time": "8am"
}
"time_slot_2":{"date":
"11/05/2021",
"time": "10am"
}
"time_slot_3":{"date":
"12/05/2021",
"time": "11am"}}
```

Figure 3.1

[Total : 25 marks]

Q5. SmartParking is a real-time mobile app that helps users find any available parking spot on a specific level of a shopping mall. To send and receive data between the app and the server, this app must establish a socket connection using TCP to **smart.parking.server1.my:8282**.

(a) The app sends the user's location in terms of the current floor level to the server as shown in the sample JSON data in Figure 4.1. The server responds to the app with the data of the listing of carpark number, with "true" indicates available and "false" indicates not available, in JSON data as shown in the sample in Figure 4.2.

Create a code segment that opens a TCP connection to the webserver URL **smart.parking.server1.my:8282** and uses the WebSocket API to send the data shown in Figure 4.1 to the server. Display a Toast to inform user that connection was made.

When the response data shown in Figure 4.2 is received, navigate the user to another screen named **Details** and pass the received data as a route parameter. Otherwise, an alert with an appropriate error message should be displayed.

If an error occurs while attempting to connect to and communicate with the web server, an alert with an appropriate error message should be displayed.

(15 marks)

Q5. (a) (Continued)

```
{
"floor_level":2
}
```

Figure 4.1

```
{
"Car_park_no_1":true
"Car_park_no_2":true
"Car_park_no_3":true
"Car_park_no_4":false
"Car_park_no_5":true
...
...
"Car_park_no_14":false
"Car_park_no_15":false
"Car_park_no_16":false
}
```

Figure 4.2

(b) Users of the mobile app can also fill out a user feedback form to report any errors or omissions in the carpark listing. Figure 4.3 shows an example of user feedback data in JSON format that will be sent from the app to the server. The server responds to the app by sending the integer '1' if it is successful and '0' if it is unsuccessful. When the app receives a response, it displays an alert with the appropriate message.

```
{
"name":"Jordan"
"email":"jdon@gmail.com"
"message":"Wrong carpark info
return in car park no 15"
}
```

Figure 4.3

Construct code segment to send the feedback as shown in Figure 4.3 to the server using the WebSocket URL **smart.parking.server1.my:8282/feedback** and handle the response accordingly. Assume that the data in Figure 4.3 is already set in a variable named **userFeedback**

If an error occurs while attempting to connect to and communicate with the web server, an alert with an appropriate error message should be displayed.

(10 marks)

[Total : 25 marks]